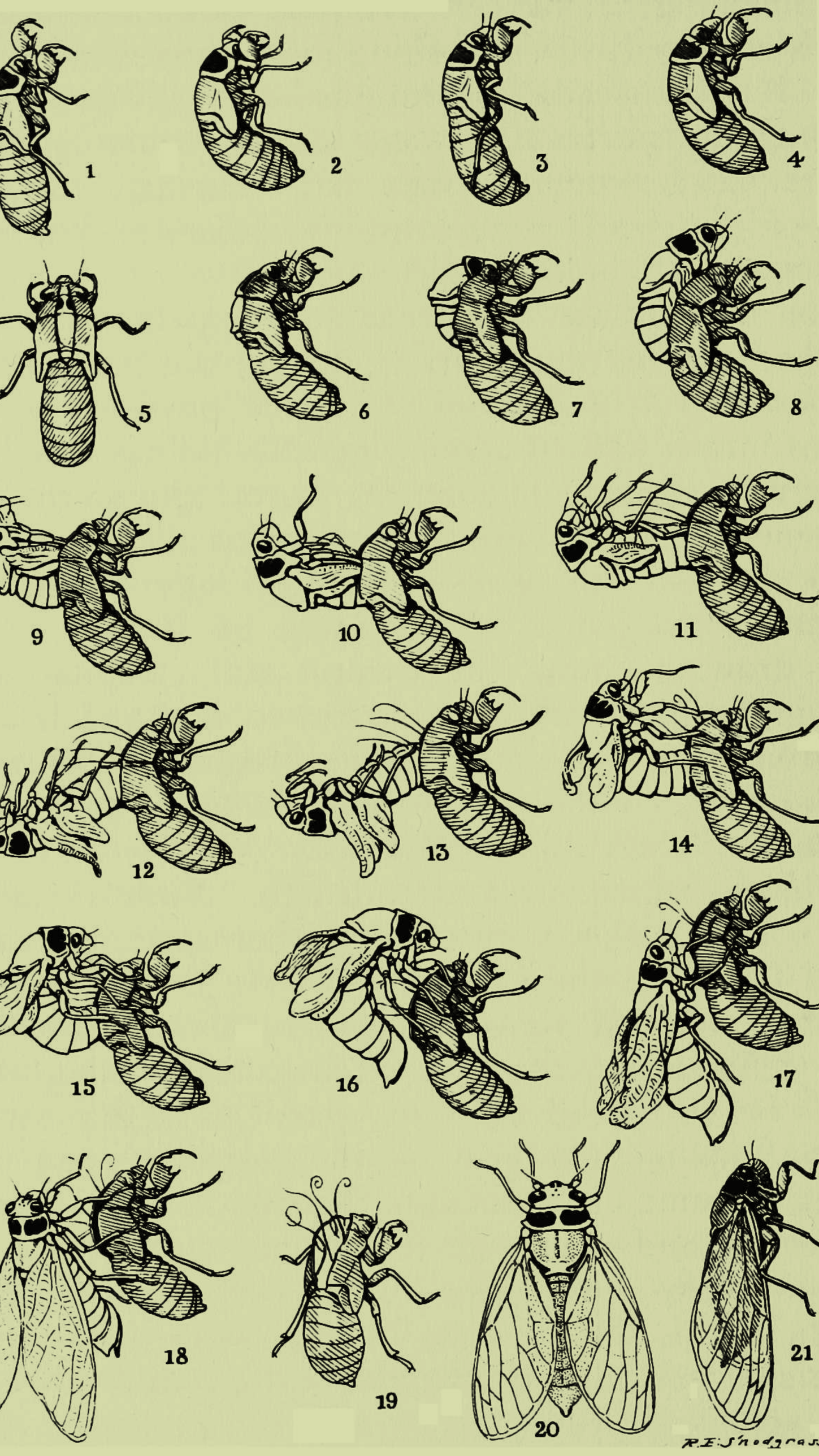
Emergent Resonances: Ultimate audiovisual experience. As close as one gets to being a cicada

Ilya Duganov and Mike Anisimov

2025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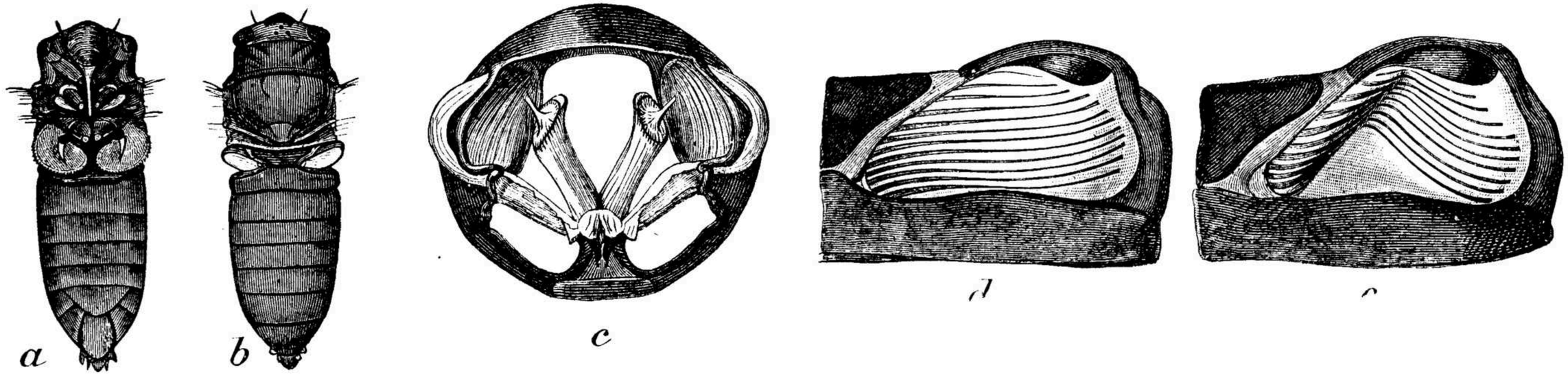
R.E. Snodgrass

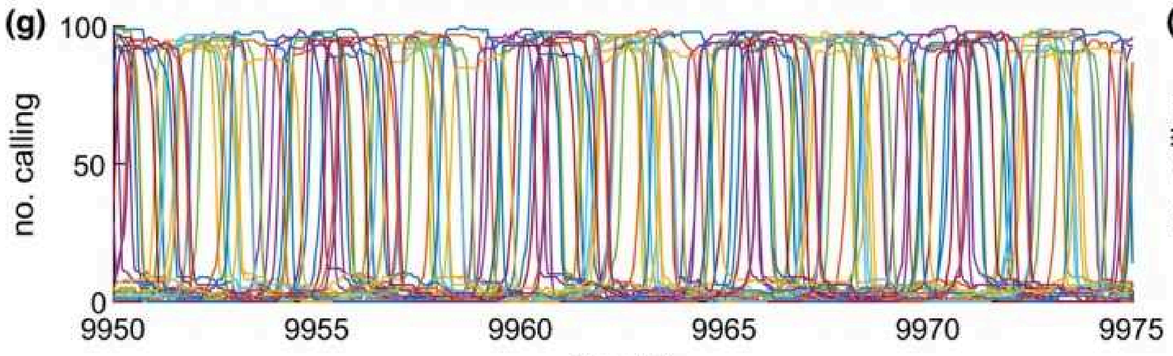
Emergent Resonances is an interactive audiovisual work inspired by cicada choruses. Participants become part of a decentralized sonic organism whose coherence arises through mutual attunement. The piece explores “mutual benefit” as emergent resonance — a collective negotiation of rhythm and relation across human, animal, and technological bodies.

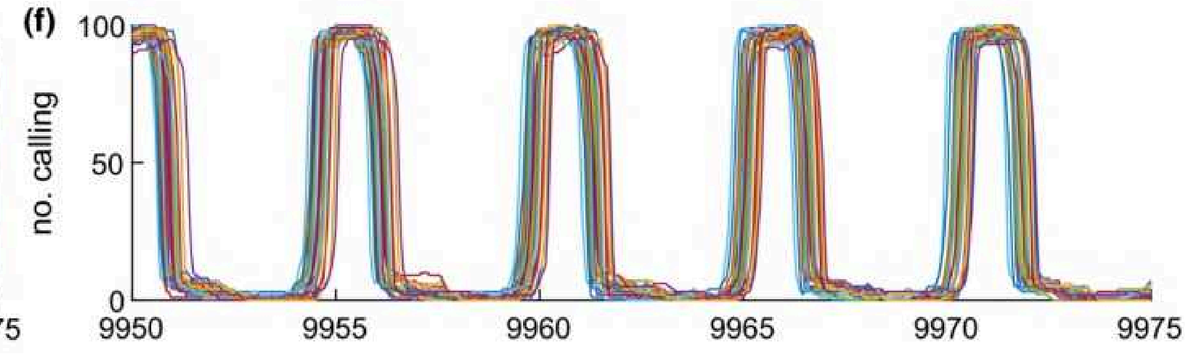
Preface → Concept → Overview → Live Prototype → Exhibition setup

Cicadas synchronise buzzing of vibrating membranes on their bodies. Groups create choruses that rise and fade in repeating cycles.



This collective behaviour is an adaptive strategy: by merging their sounds into one overwhelming wave, cicadas confuse and deter predators, which perceive the swarm as a single, powerful organism.





Participants follow a link to the web app

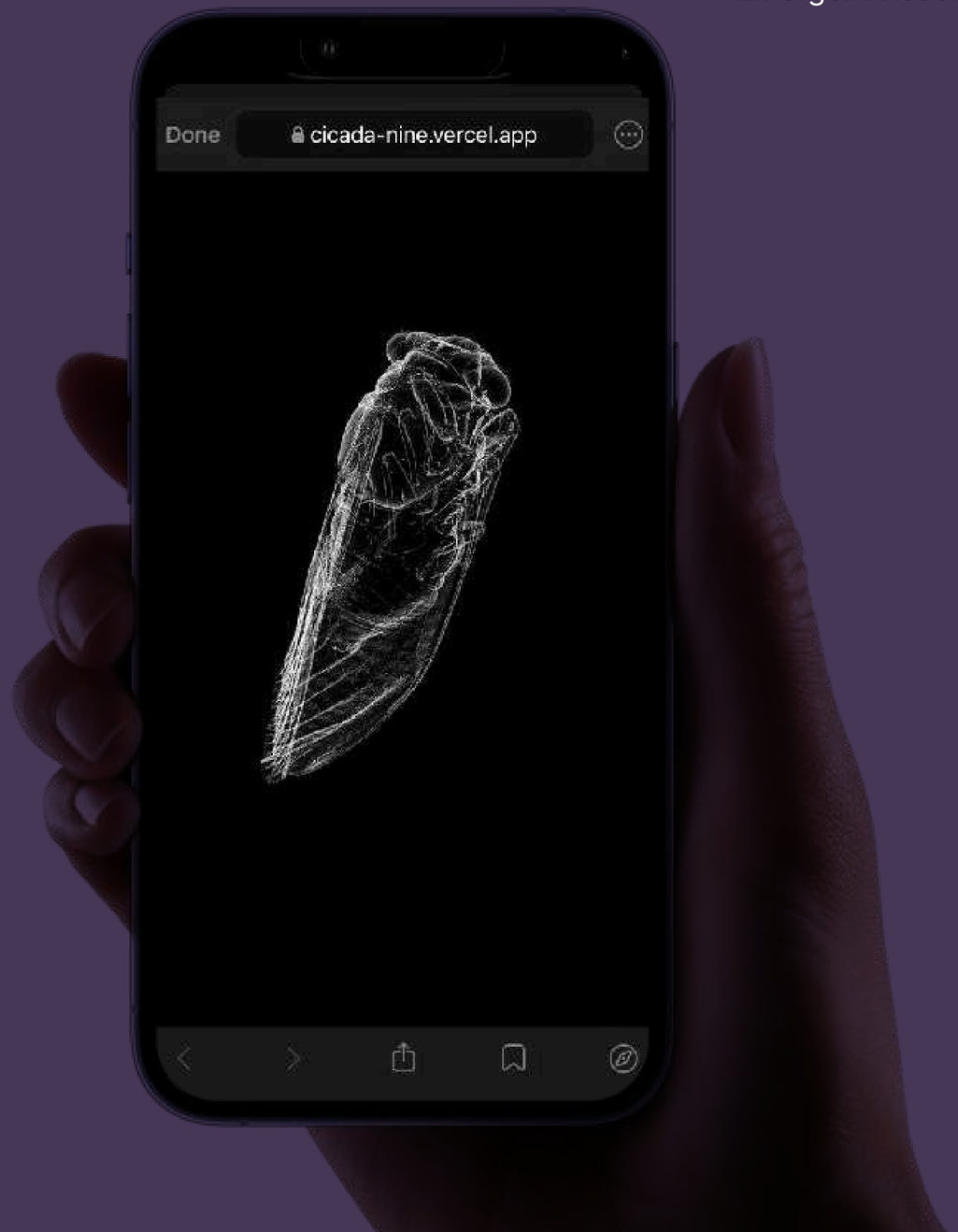
Tapping the touchscreen triggers cicada signing simulation, and cycles through frequencies.

Screen flickering is synchronised with sound amplitude providing visual feedback.

Proof of concept prototype

cicada-nine.vercel.app

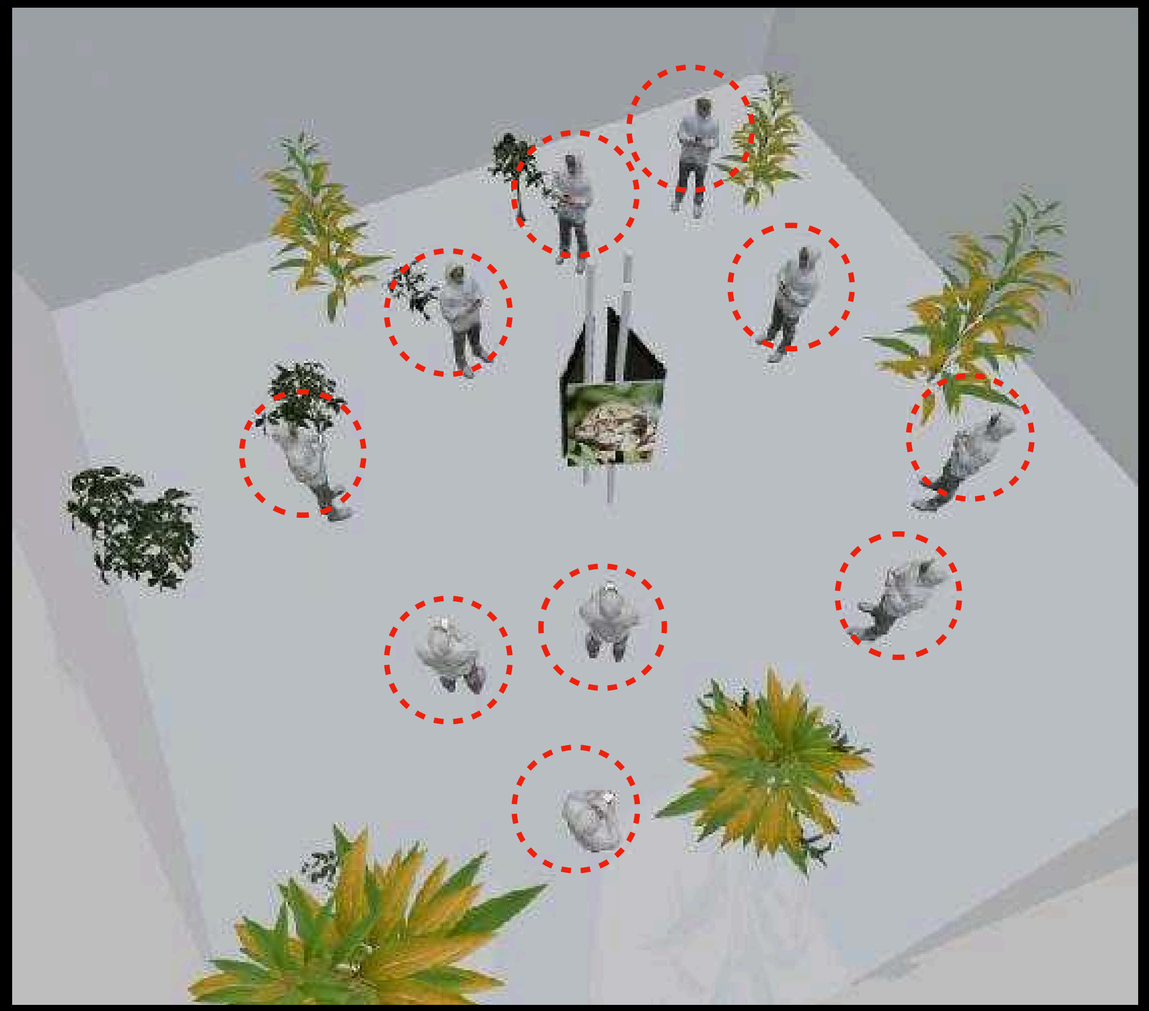
Turn off silent mode and tap the screen

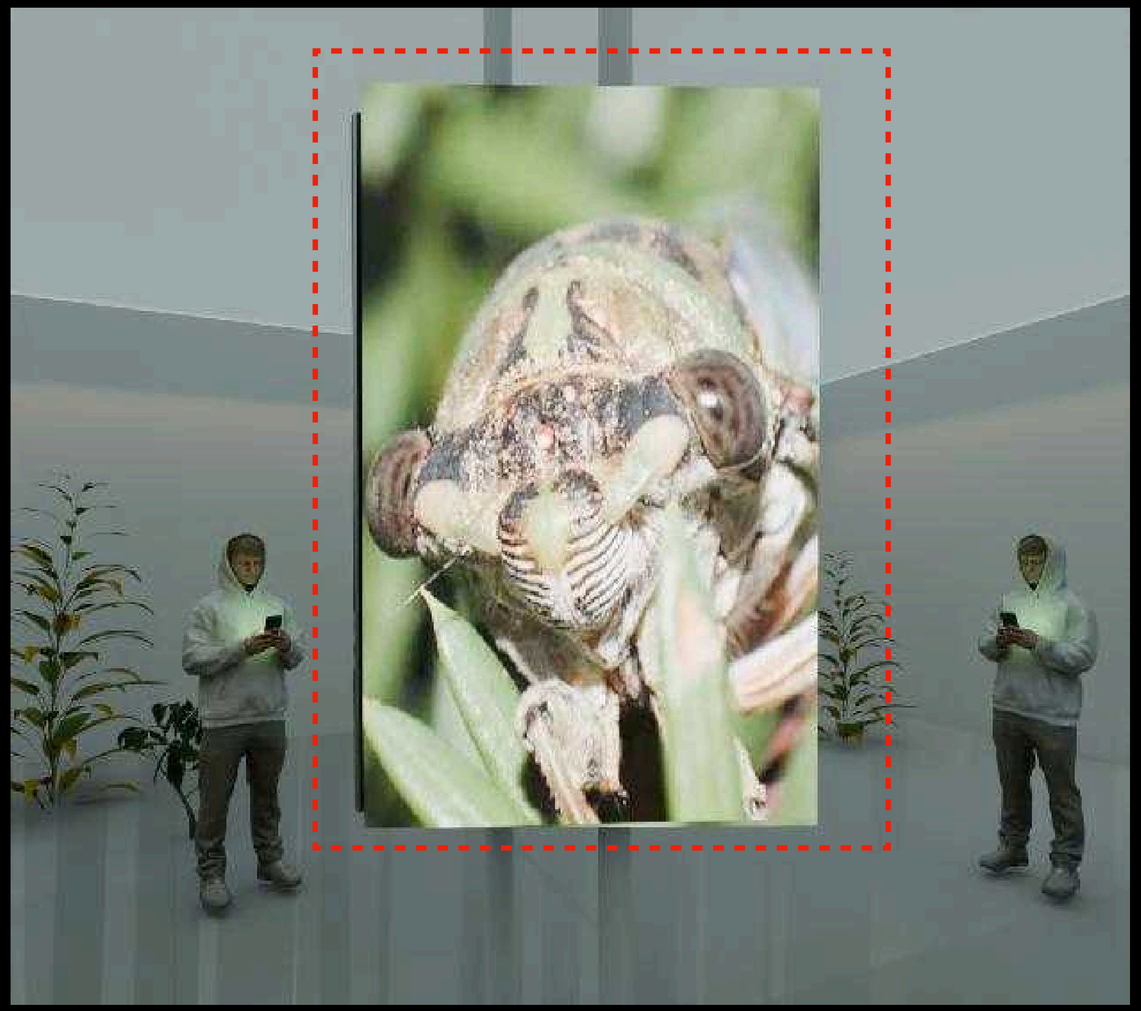


Gallery space designed for performance for up to 30 participants.

Dimmed lights preferred to deliver immersive experience and focus participants on sound of each other.

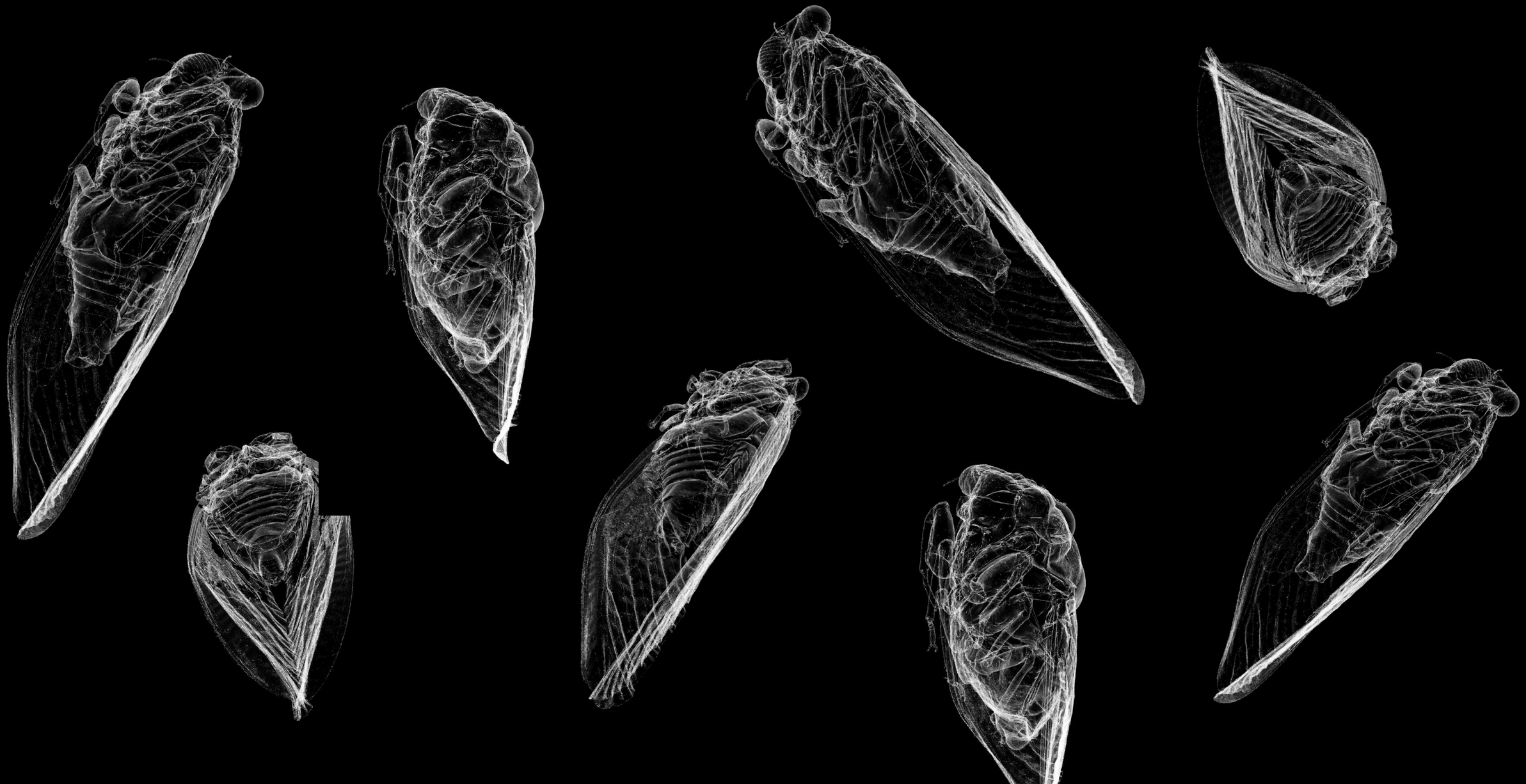
Participants mobile devices form a multichannel sound-space and engaging participatory experience.





Central piece is a cicada trainer, telling the rules and a brief story of different insect collective behaviour models.

After mr. Cicada briefing, the lights go off and the main part of the experience begins.



Thank you

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